

Problem 4

Problem 4 (CO3): Derivations, Parse Trees and Ambiguity (10 points)

Let $\Sigma = \{0, 1\}$. Consider the following grammar over Σ .

$$S \rightarrow 0S0 \mid 1S1 \mid A$$

$$A \rightarrow 00A \mid 01A \mid 10A \mid 11A \mid \varepsilon$$

- (a) **Give** a leftmost derivation for the string 01101010. (3 points)
- (b) **Draw** the parse tree corresponding to the derivation you gave in (a). (2 points)
- (c) **Demonstrate** that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the given string in (a). (4 points)
- (d) How many four-letter strings will have exactly one parse tree in the given grammar? (1 point)

Solutions:

There are three parse trees. The following table will help you to check Questions: a,b,c.

$ \begin{array}{c} S \\ \\ A \\ \backslash \\ 01A \\ \\ 10A \\ \\ 10A \\ \\ 10A \\ \\ \varepsilon \end{array} $	$ \begin{array}{l} S \\ \Rightarrow A \\ \Rightarrow 01A \\ \Rightarrow 0110A \\ \Rightarrow 011010A \\ \Rightarrow 01101010A \\ \Rightarrow 01101010 \end{array} $
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S 0 S 0 A 1 1 A 0 1 A 0 1 A ε	S ⇒ 0S0 ⇒ 0A0 ⇒ 011A0 ⇒ 01101A0 ⇒ 0110101A0 ⇒ 01101010
S 0 S 0 1 S 1 A 1 0 A 1 0 A ε	S ⇒ 0S0 ⇒ 01S10 ⇒ 01A10 ⇒ 0110A10 ⇒ 011010A10 ⇒ 01101010

Question (d): **8 strings**

(0001, 0011, 0101, 0111, 1000, 1010, 1100, 1110) → **Students don't need to write these strings**

You need to write only 1 string.